

60 > What do you need?

```
• from models import DiffusionModel
•
• # Set up device (GPU if available, otherwise CPU)
• device = torch.device('cuda' if torch.cuda.is_available()
else 'cpu')
•
• # Set up hyperparameters
• batch_size = 16
• learning_rate = 1e-4
• num_steps = 10000
• num_channels = 3
• image_size = 256
•
• # Set up data transforms
• transform = transforms.Compose([
•     transforms.Resize((image_size, image_size)),
•     transforms.ToTensor(),
•     transforms.Normalize(mean=[0.5, 0.5, 0.5], std=[0.5,
0.5, 0.5])
• ])
•
• # Load dataset
• dataset = YourDatasetHere(transform=transform)
• loader = DataLoader(dataset, batch_size=batch_size,
shuffle=True)
•
• # Set up diffusion model
• model = DiffusionModel(num_channels=num_channels).
to(device)
•
• # Set up optimizer
• optimizer = optim.Adam(model.parameters(), lr=learning_
rate)
•
• # Set up loss function (e.g., mean squared error)
• loss_fn = nn.MSELoss()
•
• # Train model
• model.train()
• for step in tqdm(range(num_steps)):
•     # Get batch of data
```